Project Report: AI Vacuum World Agent in Python

Introduction: This project implements a simple AI vacuum cleaner agent that operates in a two-room environment. The goal is for the agent to clean both rooms by perceiving the state of each room and acting accordingly. This project demonstrates core AI concepts such as state space representation, reflex agents, and environment modeling

Problem Statement: Create an intelligent agent capable of cleaning a two-room environment. The agent must decide when to clean and when to move between rooms based on sensed information about dirt presence and its location

Objectives:

Model the vacuum world environment with distinct states representing room cleanliness and vacuum location.

Develop a reflex agent that chooses actions based on the current environment state: clean if dirty, move otherwise.

Simulate and visualize the agent's cleaning process step-by-step.

Methodology:

Represent each state with the vacuum’s location (Room A or

B) and cleanliness status of both rooms.

Define possible actions: clean, move_left, move_right.Implement environment and agent classes in Python.

The agent chooses actions using simple condition checks reflecting a reflex agent.

Simulate until both rooms are clean

Implementation

The project is implemented in a single Python file vacuum_world.py.

The Vacuum World class handles environment state and transitions.

The Reflex Vacuum Agent class decides actions based on room status.

The script runs a loop simulating steps until the environment is clean, printing the agent’s decisions.

Results:

The vacuum agent successfully cleans both rooms in a finite number of steps

The console output shows agent location, room states, and actions clearly at each step.

```python
class VacuumWorld:
    def __init__(self, location, room_A_dirty, room_B_dirty):
        self.location = location  # 'A' or 'B'
        self.room_A_dirty = room_A_dirty  # True or False
        self.room_B_dirty = room_B_dirty  # True or False

    def status(self):
        return (self.location, self.room_A_dirty, self.room_B_dirty)

    def is_goal(self):
        return not self.room_A_dirty and not self.room_B_dirty

    def move_left(self):
        if self.location == 'B':
            self.location = 'A'

    def move_right(self):
        if self.location == 'A':
            self.location = 'B'

    def clean(self):
        if self.location == 'A':
            self.room_A_dirty = False
        elif self.location == 'B':
            self.room_B_dirty = False
```

```python
class ReflexVacuumAgent:
    def act(self, world: VacuumWorld):
        location, room_A_dirty, room_B_dirty = world.status()
        if (location == 'A' and room_A_dirty) or (location == 'B' and room_B_dirty):
            return 'clean'
        elif location == 'A':
            return 'move_right'
        elif location == 'B':
            return 'move_left'


# Example usage:
world = VacuumWorld(location='A', room_A_dirty=True, room_B_dirty=True)
agent = ReflexVacuumAgent()

steps = 0
while not world.is_goal():
    action = agent.act(world)
    print(f"Step {steps}: Location={world.location}, Room A Dirty={world.room_A_dirty}, Room B Dirty={world.room_B_dirty}, Action={action}")
    if action == 'clean':
        world.clean()
    elif action == 'move_left':
        world.move_left()
    elif action == 'move_right':
        world.move_right()
    steps += 1

print(f"All rooms are clean after {steps} steps.")
```

Step 0: Location=A, Room A Dirty=True, Room B Dirty=True, Action=clean

Step 1: Location=A, Room A Dirty=False, Room B Dirty=True, Action=move_right

Step 2: Location=B, Room A Dirty=False, Room B Dirty=True, Action=clean

All rooms are clean after 3 steps.